

A Robust and Explainable Lightweight Deep Learning Framework for Cross-Dataset Plant Disease Detection under Real-World Conditions

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Abstract—Accurate and reliable plant disease detection is important for supporting early disease identification and improving agricultural decision-making, particularly under real-world conditions. However, deep learning models often achieve high performance on controlled datasets while exhibiting substantial performance degradation when evaluated on images collected from different real-world domains. This study presents a lightweight and explainable deep learning framework for cross-dataset plant disease detection under real-world conditions. A MobileNetV3-Small model was trained on the PlantVillage dataset and evaluated on the completely unseen PlantDoc dataset to investigate cross-dataset generalization. The model was trained using 43,444 PlantVillage training images and validated using 10,861 images across 38 disease classes. The proposed model achieved a validation accuracy of 99.71% on PlantVillage. However, cross-dataset evaluation on PlantDoc resulted in an accuracy of 17.18%, with weighted precision, weighted recall, and weighted F1-score of 0.1978, 0.1718, and 0.1528, respectively. This represents an accuracy drop of 82.53 percentage points, highlighting the substantial domain shift between controlled and real-world plant disease images. Robustness experiments further showed that the accuracy decreased to 15.42% under brightness variation, 13.22% under Gaussian noise, and 15.42% under blur. Test-Time Augmentation did not improve the performance, achieving an accuracy of 16.74%. Finally, Grad-CAM was employed to provide visual explanations of the model predictions and to investigate the image regions influencing the classification decisions. These findings demonstrate that high in-domain accuracy alone is insufficient for reliable real-world plant disease detection and emphasize the importance of cross-dataset evaluation, robustness assessment, and explainability in the development of lightweight deep learning systems for practical agricultural applications.

Keywords—Plant Disease Detection, MobileNetV3-Small, Cross-Dataset Generalization, Explainable AI, Grad-CAM, Robustness, Deep Learning

I. INTRODUCTION

Deep learning has significantly improved automatic plant disease detection, which enabled fast and accurate diagnosis from leaf images [4], [8]. Recent lightweight Vision Transformer models, such as PMVT and ViT-based approaches, have further supported efficient deployment on mobile and

other such devices [3], [5]. These advancements contribute to precision agriculture by assisting farmers in early disease identification and timely crop management [8].

Despite these advances, there are several challenges still limiting the practical deployment of existing plant disease detection systems. Many models are trained and tested on a single dataset collected under controlled conditions, resulting in poor general results when tested on unseen datasets or real-world field images. Furthermore, many deep learning models operate as blackbox systems, providing predictions without explaining the reasons behind them, which reduces user trust and practical adoption [6], [11]. In addition, computational complexity remains a major obstacle for deploying accurate models on resource-constrained mobile devices [10].

Although many lightweight CNN and Vision Transformer models have been proposed, cross-dataset robustness remains a major challenge in plant disease detection [8],[10],[11]. Most existing studies test their models on a single dataset under controlled conditions which limit their performance in real-world environments [2],[3],[5]. Furthermore, only a limited number of studies integrate Explainable Artificial Intelligence (XAI) to provide transparent and interpretable predictions, which reduces user trust and practical adoption [1], [7].

Therefore, this research proposes a robust and explainable lightweight deep learning framework for cross-dataset plant disease detection under real-world conditions. The proposed framework aims to combine lightweight Vision Transformer architectures with Explainable Artificial Intelligence (XAI) techniques to improve classification accuracy, model transparency, and computational efficiency [3], [7]. The proposed model will be evaluated on multiple publicly available plant disease datasets to investigate its robustness across different environmental conditions.

The main contributions of this study are as follows:

- Developed a robust and lightweight deep learning framework for accurate plant disease detection under real-world conditions.

- Improved cross-dataset robustness by testing the proposed model on multiple publicly available plant disease datasets.
- Integrated Explainable Artificial Intelligence (XAI) using Grad-CAM to provide transparent visual explanations for model predictions.
- Lightweight architecture combination to get high classification accuracy with low computational complexity for mobile and other edge device deployment.

II. RELATED WORK

Recent advances in deep learning have significantly improved plant disease detection by enabling accurate classification of diseased leaf images. Lightweight architectures and Vision Transformer-based models, such as PMVT, ViT-SmartAgri, and AppViT, have demonstrated promising performance while reducing computational complexity for mobile and edge deployment [3], [5], [10]. However, these studies mainly focus on classification accuracy and lightweight design rather than cross-dataset robustness.

Several studies have further enhanced disease recognition by incorporating advanced feature extraction techniques. Swin Transformer-based architectures, lightweight MobileViT networks, and hybrid EfficientNetV2–Swin Transformer models have improved feature representation and classification performance under specific experimental settings [6], [9], [11]. Nevertheless, most of these approaches are evaluated on individual datasets collected under controlled conditions, limiting their generalization to unseen datasets and real-world agricultural environments.

Recent research has also explored Explainable Artificial Intelligence (XAI) to improve the transparency of deep learning models. Methods such as Grad-CAM and other visualization techniques have been integrated into lightweight disease detection frameworks to provide interpretable predictions and enhance user confidence [1], [7]. Furthermore, existing survey and bibliometric studies have summarized recent progress, available datasets, lightweight architectures, and the remaining challenges in plant disease detection [4], [8]. However, these studies highlight that explainability, robustness, and practical deployment remain open research challenges.

Therefore, a research gap still exists in developing a lightweight and explainable deep learning framework that performs consistently across multiple plant disease datasets under real-world conditions. To address this limitation, this study proposes a robust lightweight framework integrated with Explainable AI techniques and evaluates its performance across multiple publicly available datasets. A comparison of related works and the contributions of the proposed study is presented in Table I.

III. DATASET DESCRIPTION

The experiment was conducted using two publicly available plant disease image datasets, namely PlantVillage and PlantDoc. The PlantVillage dataset was used as the primary training and validation dataset, while the PlantDoc dataset was

TABLE I
COMPARISON SUMMARY OF RELATED WORKS AND CONTRIBUTIONS OF OUR STUDY

Paper	Lightweight	Explainable AI	Cross-Dataset
[5]	✓	×	×
[3]	✓	×	×
[10]	✓	×	×
[8]	×	×	×
[6]	×	×	×
[11]	✓	×	×
[9]	×	×	×
[7]	✓	✓	×
[2]	✓	×	×
[1]	×	✓	×
Proposed Work	✓	✓	✓

kept completely unseen during model training and used as an external test dataset for evaluating cross-dataset generalization.

The PlantVillage dataset was obtained from the Kaggle repository and contains approximately 54,305 images of healthy and diseased plant leaves distributed across 38 classes. The dataset covers 14 crop species, including apple, blueberry, cherry, grape, orange, peach, pepper, potato, raspberry, soybean, squash, strawberry, and tomato. It contains multiple plant disease categories as well as healthy leaf classes. The images were primarily collected under controlled conditions, where individual leaves are generally presented against relatively simple backgrounds. The dataset is organized into separate training and validation directories, with 38 class directories in each. Due to its large number of labeled images and well-defined class structure, PlantVillage was selected as the primary dataset for training the deep learning models.

The PlantDoc dataset was obtained from the Kaggle repository and is based on the original PlantDoc dataset introduced for visual plant disease detection. The selected Kaggle version contains approximately 2,551 image files distributed across 27 classes, with separate training and testing directories. Unlike PlantVillage, PlantDoc contains images captured under more realistic environmental conditions. The images include variations in background complexity, illumination, camera viewpoints, leaf orientations, and natural surroundings. These characteristics make PlantDoc more challenging than PlantVillage and provide a suitable external dataset for evaluating model robustness under real-world conditions.

In this study, PlantDoc was not used during model training or model selection. Instead, it was reserved as an unseen external test dataset. The trained models were evaluated on PlantDoc after completion of the training and validation process. This experimental design allows the proposed framework to be evaluated under a distribution shift between the source training dataset and the target testing dataset.

The use of PlantVillage for model development and PlantDoc for independent evaluation provides a suitable experimental setting for investigating cross-dataset generalization. In particular, PlantVillage represents the source domain with relatively controlled image conditions, whereas PlantDoc rep-

resents a different target domain with more realistic image variations. Therefore, the performance of the trained models on PlantDoc provides an indication of their ability to generalize beyond the conditions observed during training.

A. Data Preprocessing

Several preprocessing steps were performed before training the deep learning models, including image resizing, normalization, data augmentation, dataset splitting, and class distribution analysis.

All input images were resized to 224×224 pixels to provide a consistent input resolution for the selected lightweight deep learning architectures. Pixel values were normalized using the ImageNet normalization parameters, which is appropriate for models initialized using ImageNet-pretrained weights.

Data augmentation was applied only to the training images to increase the diversity of the training data and reduce overfitting. The augmentation pipeline included transformations such as random rotation, horizontal flipping, scaling or zooming, brightness variation, and translation. These transformations were selected to expose the model to variations that may occur in real-world agricultural environments while preserving the semantic characteristics of the plant disease classes.

The PlantVillage dataset was divided into training and validation sets while maintaining the original class structure. The training subset was used for model parameter optimization, whereas the validation subset was used for monitoring model performance, selecting the best-performing model, and applying early stopping when necessary.

The PlantDoc dataset was completely excluded from the model development process. It was used only after the training process as an unseen external test dataset. This strict separation prevents information from PlantDoc from influencing model training, hyperparameter selection, or model selection and enables a genuine cross-dataset evaluation.

The class distribution of the PlantVillage training data was analyzed before model training to identify potential class imbalance. Where necessary, appropriate class-balancing strategies were considered to reduce the potential bias of the model toward classes containing a larger number of training samples.

The overall dataset configuration used in the study is summarized in Table II.

TABLE II
DATASET DISTRIBUTION OF PLANT DISEASE DETECTION DATASET

Dataset	Purpose	Images	Classes	Condition
PlantVillage	Training/Validation	54,305	38	Controlled
PlantDoc	Testing	2,551	27	Real-world

The PlantVillage dataset contains approximately 54,305 images organized into 38 classes, while the selected PlantDoc Kaggle repository contains approximately 2,551 image files organized into 27 classes. Since the two datasets do not contain identical class structures, only the compatible classes between

the source and target datasets are considered during cross-dataset evaluation. The final number of images and compatible classes used in the experimental evaluation will be reported based on the actual downloaded dataset and the class-mapping procedure applied in the implementation.

IV. METHODOLOGY

This study proposes a lightweight, robust, and explainable deep learning framework for plant disease detection under real-world agricultural conditions. The proposed framework aims to improve cross-dataset generalization while maintaining low computational complexity for deployment on resource-constrained devices. The overall workflow, including dataset collection, preprocessing, model training, model selection, cross-dataset evaluation, explainable AI, and final performance analysis, is illustrated in Fig. 1.

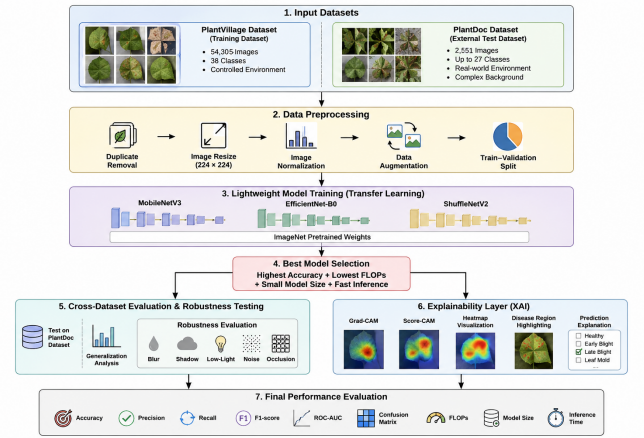


Fig. 1. System architecture of the proposed plant disease detection framework

Two publicly available datasets, PlantVillage and PlantDoc, were used in this study. PlantVillage was utilized for model training due to its large collection of controlled leaf images, while PlantDoc was used as an unseen external test dataset to evaluate real-world generalization under complex field conditions.

Before training, images were preprocessed by removing duplicates, resizing them to 224×224 pixels, and normalizing pixel values. Data augmentation techniques, including rotation, flipping, zooming, translation, and brightness adjustment, were applied to improve model robustness. The PlantVillage dataset was divided into training and validation sets, whereas PlantDoc was reserved only for external evaluation.

Three lightweight transfer learning models, MobileNetV3, EfficientNet-B0, and ShuffleNetV2, were trained using ImageNet pretrained weights and fine-tuned on the PlantVillage dataset. The models were compared based on accuracy, precision, recall, F1-score, model size, inference time, FLOPs, and parameter count. The model achieving the best balance between performance and computational efficiency was selected as the final model.

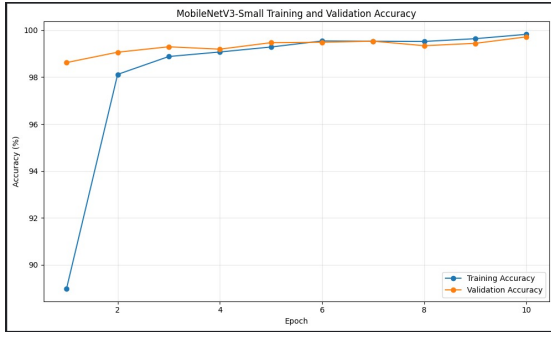


Fig. 2. Training and validation accuracy of MobileNetV3-Small on PlantVillage.

The selected model was further evaluated using cross-dataset testing on PlantDoc and robustness analysis under challenging conditions, including blur, noise, low illumination, shadow, and occlusion. To improve model transparency, Explainable AI techniques, including Grad-CAM and Score-CAM, were incorporated to generate visual explanations highlighting important disease-related regions.

Finally, the proposed framework was evaluated using classification metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix, along with computational metrics including inference time, model size, FLOPs, and parameter count. Error analysis was also performed to assess model limitations and practical applicability for mobile and edge-based plant disease detection.

V. EXPERIMENTS AND RESULTS

The proposed lightweight deep learning framework was evaluated using PlantVillage as the training and validation dataset and PlantDoc as a completely unseen external test dataset. A MobileNetV3-Small model was trained using 43,444 PlantVillage training images and validated using 10,861 images across 38 disease classes. The input images were resized to 224×224 pixels, and the model was trained for 10 epochs.

The training and validation accuracy of the MobileNetV3-Small model is shown in Fig. 2. The model achieved a final training accuracy of 99.82% and a validation accuracy of 99.71%. The best validation accuracy of 99.71% was obtained at the 10th epoch.

The corresponding training and validation loss curves are presented in Fig. 3. The training loss decreased throughout the training process, while the validation loss reached 0.0084 at the final epoch. These results demonstrate that the model learned the visual patterns present in the PlantVillage dataset effectively.

A. Cross-Dataset Evaluation

To evaluate the cross-dataset generalization capability of the proposed framework, the trained model was evaluated on the PlantDoc dataset, which was kept completely unseen during model training and validation. PlantDoc contained 236 test images. Due to differences between the PlantDoc and



Fig. 3. Training and validation loss of MobileNetV3-Small on PlantVillage.

TABLE III
PLANTDOC CROSS-DATASET EVALUATION RESULTS.

Metric	Value
Total PlantDoc Images	236
Evaluated Images	227
Skipped Images	9
Correct Predictions	39
Incorrect Predictions	188
Accuracy	17.18%
Weighted Precision	0.1978
Weighted Recall	0.1718
Weighted F1-score	0.1528
Macro F1-score	0.1425

PlantVillage label spaces, 26 classes could be mapped to the corresponding PlantVillage classes. Therefore, 227 images were evaluated, while 9 images were skipped because their classes could not be mapped.

The cross-dataset evaluation results are presented in Table III. The model correctly classified 39 out of 227 evaluated images, achieving an accuracy of 17.18%. The weighted precision, weighted recall, and weighted F1-score were 0.1978, 0.1718, and 0.1528, respectively. The macro F1-score was 0.1425.

The difference between the PlantVillage validation performance and the PlantDoc cross-dataset performance highlights the domain-shift problem in plant disease recognition. Although the model achieved 99.71% validation accuracy on PlantVillage, its accuracy decreased to 17.18% on the unseen PlantDoc dataset. This represents an accuracy drop of 82.53 percentage points. The result indicates that high performance under controlled image conditions does not necessarily guarantee strong generalization to real-world field images.

B. Confusion Matrix Analysis

The class-wise prediction behavior of the model on the PlantDoc dataset is illustrated using the confusion matrix in Fig. 4. The confusion matrix contains predictions for the 26 mapped classes.

The confusion matrix shows that the model performs differently across plant disease classes. Some classes achieved relatively higher recall, while several classes obtained very low or zero recall. For example, Pepper bell healthy achieved

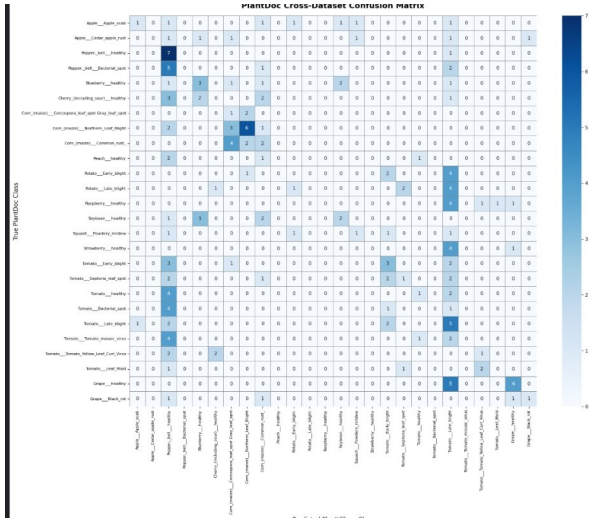


Fig. 4. PlantDoc cross-dataset confusion matrix for the 26 mapped classes.

TABLE IV
ROBUSTNESS EVALUATION UNDER DIFFERENT CONDITIONS.

Condition	Accuracy
Original	17.18%
Brightness	15.42%
Gaussian Noise	13.22%
Blur	15.42%

a recall of 0.88, while Corn Northern Leaf Blight and Tomato Late Blight achieved recalls of 0.50. In contrast, several classes including Apple Cedar Apple Rust, Cherry healthy, Peach healthy, Potato Early Blight, Potato Late Blight, Raspberry healthy, and Strawberry healthy obtained zero recall. These results further demonstrate the difficulty of transferring knowledge learned from PlantVillage to the real-world PlantDoc domain.

C. Robustness Evaluation

The robustness of the trained model was evaluated under different image perturbation conditions. In addition to the original PlantDoc images, brightness variation, Gaussian noise, and blur were applied to the evaluated images.

The original PlantDoc images produced an accuracy of 17.18%. Under brightness variation, the accuracy decreased to 15.42%. Gaussian noise produced the largest degradation, reducing the accuracy to 13.22%. Blur also reduced the accuracy to 15.42%. These results indicate that image-quality and environmental variations can further affect the prediction performance of the model.

D. Test-Time Augmentation Analysis

Test-Time Augmentation (TTA) was additionally evaluated to determine whether predictions could be improved by aggregating predictions from multiple transformed versions of the same image. The original evaluation achieved an accuracy of 17.18%, whereas TTA achieved 16.74%. Therefore, TTA

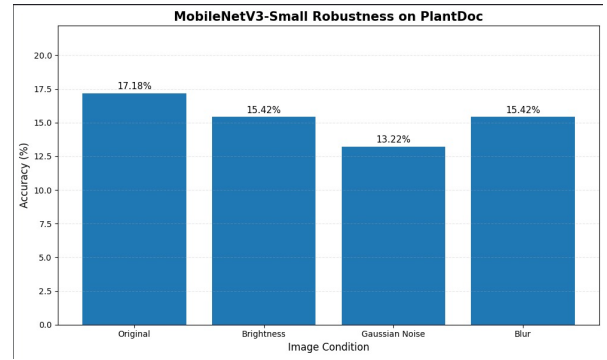


Fig. 5. Representative Grad-CAM visualizations of PlantDoc images.

resulted in a decrease of 0.44 percentage points and did not improve the cross-dataset performance of the model.

E. Grad-CAM Explanations

Grad-CAM was used to provide visual explanations of the MobileNetV3-Small predictions. The objective of this analysis was to identify the image regions that contributed most strongly to the model's predictions. This visualization provides an interpretable representation of the model's decision-making process and helps investigate whether the model focuses on relevant leaf regions.

Representative Grad-CAM visualizations generated from the PlantDoc images are presented in Fig. 5. The highlighted regions provide visual information about the areas receiving greater attention from the trained network.

Overall, the experimental results demonstrate a substantial difference between in-domain performance and cross-dataset generalization. The MobileNetV3-Small model achieved 99.71% validation accuracy on PlantVillage, whereas its performance on the unseen PlantDoc dataset was 17.18%. The robustness evaluation further demonstrated performance degradation under brightness, Gaussian noise, and blur perturbations. The TTA experiment also failed to improve the cross-dataset accuracy. Finally, the Grad-CAM analysis provides an explainability component by visualizing the image regions influencing the model's predictions.

VI. CONCLUSION

This study presented a lightweight and explainable deep learning framework for cross-dataset plant disease detection under real-world conditions. The proposed framework used MobileNetV3-Small as the primary lightweight classification model, with PlantVillage used for training and validation and PlantDoc kept completely unseen for external cross-dataset evaluation. The framework was designed to investigate not only in-domain classification performance but also cross-dataset generalization, robustness to image perturbations, and model interpretability.

The experimental results showed that MobileNetV3-Small achieved a validation accuracy of 99.71% on the PlantVillage dataset, demonstrating strong performance under the controlled image conditions of the training dataset. However,

when evaluated on the unseen PlantDoc dataset, the model achieved an accuracy of only 17.18%, with a weighted F1-score of 0.1528 and a macro F1-score of 0.1425. The substantial difference between the PlantVillage validation accuracy and PlantDoc performance indicates a pronounced domain shift between controlled laboratory-style images and real-world field images.

The robustness evaluation further demonstrated that environmental and image-quality variations can negatively affect model performance. The accuracy decreased from 17.18% on the original PlantDoc images to 15.42% under brightness variation, 13.22% under Gaussian noise, and 15.42% under blur. Test-Time Augmentation also did not improve the cross-dataset performance, producing an accuracy of 16.74% compared with 17.18% for the original evaluation. These findings demonstrate that achieving high in-domain accuracy alone is insufficient for reliable plant disease recognition under real-world conditions.

The Grad-CAM analysis provided an additional explainability component by visualizing the image regions that influenced the model's predictions. This analysis helps investigate the model's decision-making behavior and provides visual evidence for assessing whether the network attends to meaningful disease-related regions of plant leaves.

Overall, the findings highlight the importance of evaluating plant disease classification models beyond conventional validation accuracy. The substantial domain gap, sensitivity to image perturbations, and limited cross-dataset performance demonstrate the need for domain-oriented training strategies and robust lightweight models for practical deployment. The proposed framework establishes a baseline for future improvements in cross-dataset generalization, robustness, and explainable plant disease detection, particularly for resource-constrained and real-world agricultural applications.

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